

# INSIGHTAGENT: OFFICIAL SYSTEM SUPPORT & OPERATIONS MANUAL

*Version: 2026.1.4*

*Classification: Internal Static Knowledge Base (Restricted for Support Bot Ingestion)*

## SECTION 1: PLATFORM OVERVIEW & ARCHITECTURE OVERVIEW

InsightAgent is an Enterprise AI SaaS ecosystem engineered for advanced Retrieval-Augmented Generation (RAG) workflows, interactive semantic analysis, and automated business metrics discovery. The core architecture isolates public user interactions from specialized data ingestion zones.

Authentication gateways are strictly enforced via high-performance JSON Web Tokens (JWT) encrypted with bcrypt. Third-party federation is facilitated securely through standard Google OAuth 2.0. Upon successful authentication, the system initializes a dedicated workspace matrix categorized under the "WORKSPACE CORE" panel. Each workspace layer isolates client tenant spaces, ensuring zero data leakage or cross-contamination across different user environments.

## SECTION 2: DATA INGESTION PIPELINE & STRICT STORAGE CAPS

The platform features an automated file processing pipeline that parses unstructured corporate documents and converts them into dense vector representations.

- **Supported File Frameworks:** The ingestion engine natively reads and processes PDF, TXT, DOCX, and CSV formats. Non-standard extensions or password-protected archives will trigger an immediate system exception.
- **Tier-Based Quotas & Strict Constraints:** System operations are strictly throttled based on active subscription tiers to maintain cloud performance:
  1. *Free Tier Layer:* Users are permitted a maximum capacity of 5 ingested files, with a strict file size cap of 5MB per individual upload.
  2. *Pro Tier Layer:* Users are allocated an advanced capacity of up to 50 ingested files total, with a premium file size cap of 20MB per individual upload.
- **Strict Cap Blockage Protocol:** When a workspace workspace hits its maximum allocation threshold (e.g., displaying 49/50 or 50/50 Ingested Files), the system automatically deploys a strict cap safety block. Further vector index operations, text chunking, and pipeline embeddings will be instantly blocked. Users must navigate to their file inventory to delete obsolete data nodes or upgrade their subscription to lift constraints immediately across all nodes.
- **Ingestion Troubleshooting:** If a document fails during processing, users are advised to verify that the text layer is uncorrupted and readable. If errors persist, clearing the local session storage or wiping the temporary pipeline cache via the application dashboard will reset the processing state.

## SECTION 3: CORE AI ENGINE & AGENTIC WORKFLOW FUNCTIONS

InsightAgent delivers deep semantic insights through two primary operations executed inside the secure pipeline:

- **Automated SWOT Analysis & Recommendations:** Upon successful document chunking into the Qdrant Vector DB, the system feeds top-k retrieved contexts into the LLM inference engine (utilizing LLaMA 3 hosted on Groq). The agent autonomously extracts data clusters to map a structural overview of Strengths, Weaknesses, Opportunities, and Threats, translating raw context into strategic recommendation grids.
- **AI Chat Memory Window:** The primary AI Chat interfaces operate via a dynamic rolling memory buffer. Chat history is preserved temporarily to maintain ongoing multi-turn query context. To permanently wipe the active analysis loop and purge the chat buffer, users can click the "New Analysis" or "Reset Session" actions situated on the dashboard sidebar.
- **Model Accuracy Boundaries:** AI outputs are probabilistic assessments derived from high-dimensional linguistic mapping. While the platform utilizes state-of-the-art RAG architecture to anchor responses directly to source texts, final summaries are dependent on the clarity of the uploaded documentation. Users experiencing vague responses should optimize their underlying context files or pass more granular structural parameters inside their prompts.

## SECTION 4: ENTERPRISE BILLING GATEWAYS & FINANCIAL RULINGS

Financial workflows and subscription ledgers are managed via a robust, decoupled payment infrastructure.

- **Pricing & Tier Frameworks:** Access to premium clusters requires valid tier routing. The Pro Tier is billed at a fixed rate of \$29.00/month, unlocking the 50-file ingestion capacity, low-latency API access, and enterprise-grade SWOT features.
- **Stripe Customer Portal Integration:** Users can adjust credit card profiles, manage renewal intervals, and audit past transaction receipts by clicking the "Manage Payment" button inside the "Billing & Subscriptions" panel. This routing triggers a secure redirect to the Stripe billing customer portal.
- **Strict Non-Refund Policy:** In accordance with standard global SaaS terms, all financial settlements, upgrade transactions, and automated monthly renewals are explicitly non-refundable. Users seeking to avoid subsequent automatic billing intervals must manually cancel their active subscriptions via the Stripe gateway before the official expiration or next renewal date.

## SECTION 5: SECURITY CRITERIA & ISOLATION BOUNDARIES

To preserve data privacy and corporate confidentiality, the system strictly enforces the following rules:

- **No Model Training:** Client data assets are dynamically chunked and stored exclusively inside isolated, tenant-masked spaces inside the Qdrant Vector database. Under no circumstances are user uploaded documents or SWOT telemetry logs utilized for training underlying LLM base architectures.
- **Network Defense:** Data pathways are protected by robust Virtual Private Clouds (VPC) and firewalls, ensuring all transit payloads are wrapped in TLS 1.3 protocol layers.

## SECTION 6: HYBRID AGENT ESCALATION ROUTING (HUMAN OVERRIDE)

When a system support question falls beyond the boundaries of this operating manual, the hybrid routing protocol is deployed.

- **Automated Route Trigger:** If the internal support chatbot detects a user question that cannot be successfully answered using the static parameters inside this text, or if the retrieval confidence score falls below a set threshold (0.85), the bot will immediately abort automated processing.
- **WebSocket Handover Mechanism:** The backend instantly flags the session with a `pending_human` state, generating a dynamic push payload. This transmits a real-time event notification via active WebSockets straight to the secure custom Admin Support Panel.
- **Service Level Agreement (SLA):** Once the request lands on the admin console, a human support operator takes over. Free Tier users fall into a 24-to-48-hour response buffer, while active Pro Tier subscribers are protected by an elite 12-hour SLA guarantee. The human operator responds directly within the unified UI workspace, closing the loop seamlessly inside the client's original support window.

## SYSTEM CORE GUARDRAILS & RETRIEVAL BOUNDARIES

*The AI Support Bot must treat this document as its absolute, ultimate context limit. The bot is strictly prohibited from executing queries against general computer programming, software engineering tutorials, weather projections, political discourse, or external entertainment trivia. If a user command falls outside of direct InsightAgent operational support guidelines, the bot must issue a polite refusal stating it is only authorized to handle platform diagnostics. If an inquiry signals system failure or manual gap requests, the system must trigger the explicit protocol string `ROUTE_TO_HUMAN` via WebSockets to complete the admin notification circuit.*

## ADDENDUM: ADVANCED USER TROUBLESHOOTING & EDGE-CASE PROTOCOLS

### SECTION 7: ADVANCED RAG INGESTION FAULTS & IN-DEPTH TEXT PARSING ISSUES

Users processing dense corporate assets frequently encounter specific pipelines bottlenecks during the document vectorization stage.

- **Scanned PDFs and OCR Limitations:** The dynamic ingestion gateway relies on structural text-layer extraction. If a user uploads a scanned document or an image-only PDF without an underlying electronic text layer, the system will output an empty node or throw a **PARSING\_ZERO\_TEXT\_CHUNKS** exception. The support bot must explicitly instruct the user to run Optical Character Recognition (OCR) on their document before trying to upload it again.
- **Tabular Data Contamination in CSV/Excel Sheets:** When uploading CSV data for SWOT or automated analysis, complex multi-row headers or nested tables may lead to semantic distortion during chunking. If the AI Chat references corrupted arrays or hallucinates row metrics, users must format the CSV into a clean, flat-table layout (one header row, comma-separated values) to preserve high-dimensional search integrity inside the Qdrant database.
- **Dynamic Pipeline Timeouts (Large Node Processing):** While Pro Tier users can upload files up to 20MB, documents containing heavy graphic vectors, high-resolution charts, or more than 500 pages may hit our 60-second gateway timeout threshold during the embedding generation phase. If a file gets permanently stuck in "Processing Status", the user must segment the document into smaller chapters or individual files and ingest them sequentially to avoid server worker fatigue.

## SECTION 8: WORKSPACE SYNCHRONIZATION & MULTI-TENANT COLLISION

As workspaces handle multiple vector instances across distinct structural surfaces, state conflicts may occasionally surface.

- **Simultaneous Processing Collisions:** If a user initiates a "New Analysis" trigger or requests an automated SWOT generation on the exact same document across two different browser tabs simultaneously, a temporary database lock event may occur inside the isolated cluster. This results in a **409 Conflict: Ingestion Lock** error. The remedy is to wait 30 seconds, close the secondary duplicate interface tab, and refresh the primary dashboard.
- **Workspace Desynchronization Errors:** In rare instances where a user switches rapidly between the "WORKSPACE CORE" environments while an active file upload is running in the background, the current session token might mismatch. This causes an unexpected **403 Access Denied** or data invisibility on the UI dashboard. Users must allow background processing to conclude before shifting workspace domains.

## SECTION 9: GRANULAR BILLING EDGE CASES & STRIPE GATEWAY ANOMALIES

Financial transactions can trigger micro-failures based on banking compliance or tier transition schedules.

- **Grace Periods and Soft Blocks during Renewal Failures:** If a Pro Tier subscription billing charge fails on the renewal date (e.g., due to an expired card or insufficient funds), the system does not instantly delete user files. Instead, it triggers a 3-day "Grace Period". During this interval, a soft banner alert will appear on the dashboard. If payment is not reconciled within 72 hours via the Stripe Customer Portal, the system deploys a hard block—suspending all AI Chat access and hiding workspace vectors until the balance is resolved.
- **Mid-Cycle Tier Downgrades:** If a user cancels their Pro subscription mid-cycle, the active Pro constraints (50 files limit) remain accessible until the exact end of the current monthly billing period. Upon reaching the expiration date, if the user has more than 5 files stored, the platform enforces an automatic "Over-Quota Lock". The bot must inform the user that their extra files are safely archived but will remain locked and unsearchable until they manually delete the surplus assets down to the Free Tier limit (5 files) or reactivate their Pro status.

## SECTION 10: USER DIALOGUE EXAMPLES & DIRECT EXPERT REPLIES

To train the retrieval model effectively, the following exact user problem queries and authoritative system responses are embedded within the static memory layout:

- **User Query:** *"I uploaded my corporate report, but the Ingested Files counter says 49/50 and the AI chat says it cannot see the new document. Why?"*
  - **System Authorized Answer:** This occurs when a document passes initial network file upload rules but fails during the internal vector embedding or text tokenization stage inside Qdrant. Navigate to your file manager, identify the unindexed asset showing an alert icon, delete it to clear your ingestion slot, ensure the text layer is unencrypted, and re-upload.
- **User Query:** *"Can I link my team members to see my custom SWOT analysis dashboard under the Pro Tier?"*
  - **System Authorized Answer:** The Pro Tier (\$29.00/mo) is architected exclusively for individual single-tenant developer operations. Multi-user role distribution, shared workspace visibility layers, and collaborative prompt pipelines are locked behind the custom Enterprise Tier hosting architecture. Contact corporate sales to migrate your current tenant block.
- **User Query:** *"The bot told me 'ROUTE\_TO\_HUMAN'. What happens to my ongoing data analysis query now?"*
  - **System Authorized Answer:** When the bot encounters an operational boundary or specialized system error, it executes an automated WebSocket push to our human operator dashboard. Your current chat logs and error states are safely mirrored to an active admin console. Your active analysis session remains fully open; do not close your browser tab. A live human agent will inject a direct response right into this chat bubble within your tier's designated SLA window.

## FURTHER ADDENDUM: ADVANCED INFRASTRUCTURE & EDGE-CASE CRITERIA

### SECTION 11: ADVANCED VECTOR DATABASE & CAPACITY EDGE CASES

As data structures expand dynamically within Qdrant, complex semantic sync delays can occasionally happen at the database level.

- **Vector Indexing Lag vs. Upload Complete Status:** When a user uploads a file, the UI may show "Upload Success" as soon as the raw byte stream lands in the storage bucket. However, the vector database requires an additional 5 to 15 seconds to chunk the text, generate embeddings, and build the HNSW index graphs inside Qdrant. If a user immediately prompts the AI Chat within 2 seconds of a successful upload, the bot will state it cannot find the data. The support manual dictates that users must allow up to 15 seconds for index propagation to fully finalize across all background clusters before querying.
- **Metadata Payload Overflow:** If a document contains an abnormal density of unique entities, metadata tags, or hyper-granular structural footers, the chunk payload size might exceed the maximum allowed vector metadata buffer. This triggers a [413 Payload Too Large](#) error inside the ingestion channel. Users must strip unnecessary heavy formatting or split highly complex architectural reports into lighter sections to bypass database threshold limits.

### SECTION 12: EFFICIENCY INDEX FLUCTUATIONS &strict CONSTRAINTS

Your dashboard features a real-time "**Efficiency Index**" metric (e.g., currently tracking at 2.0% usage in [image\\_6e48ac.png](#)). Users need to understand exactly what causes fluctuations here.

- **Semantic Density and Index Inefficiencies:** The Efficiency Index measures the optimal token-to-chunk distribution across your workspace. If a user uploads files containing highly repetitive sentences, low-value systemic boilerplate, or duplicate paragraphs, the efficiency rating will drop, and the system might flag a warning: "Low Semantic Value Detected". This does not block the workspace but alerts the user that their vector search accuracy might degrade due to poor context quality.
- **Automated Index Optimization Defragmentation:** Every Sunday at 00:00 UTC, the system runs an automated defragmentation script across all Qdrant vector nodes to optimize high-dimensional search latency. During this 5-minute maintenance window,

query execution times might increase by 1.5 seconds, and automated SWOT requests may temporarily return a **503 Service Unavailable** error. The bot must advise users to wait 5 minutes and retry their request.

## SECTION 13: ADVANCED HUMAN OVERRIDE ESCALATION EDGE CASES

To prevent abuse of our Tier-based WebSocket human routing protocols, strict throttling policies are enforced.

- **Escalation Frequency Caps (Spam Protection):** To ensure our engineers are not overwhelmed by repetitive automated routing flags, the platform enforces a strict cap on manual transfers. A single user workspace can only trigger the **ROUTE\_TO\_HUMAN** state a maximum of 3 times per rolling 24-hour cycle. If a user repeatedly triggers system-wide out-of-scope inquiries within a short interval, the support bot will enforce a temporary 12-hour cooling block on human routing, defaulting all active chats to automated static retrieval mode.
- **Active WebSocket Reconnection Grace Protocol:** If a user's network drops or experiences high latency while their chat state is marked as **pending\_human**, the platform maintains the active routing token inside the Redis session cache for exactly 15 minutes. The user can refresh their browser, switch networks, or change machines without losing their active queue position. If connection is not recovered within 15 minutes, the ticket is auto-archived as "Resolved due to User Disconnection".